

Beamformer constant sweep

Average-best DAS, DMAS, and DMAS-D4 constants on a stratified UM-BMID cohort. Research and educational use only. Not a clinical diagnosis.

Chosen: DAS γ = 0.55, DMAS pair = 0.30, DMAS γ = 0.00, D4 exponent = 0.25. Grid: 0.00 to 1.00 step 0.05. Tie band: 0.01. Score weights: 0.70 tumor / 0.30 healthy, and 0.50 DMAS / 0.50 D4 for the shared triple.

DAS coherence gamma — full grid

Value	Average	Tumor	Healthy	Loc cm	Area	Ring	Healthy FP
0.00	0.566	0.381	1.000	5.34	0.024	1.06	0.00
0.05	0.571	0.387	1.000	5.30	0.015	1.06	0.00
0.10	0.573	0.390	1.000	5.31	0.010	1.07	0.00
0.15	0.575	0.392	1.000	5.40	0.007	1.06	0.00
0.20	0.579	0.399	1.000	4.97	0.005	1.11	0.00
0.25	0.606	0.437	1.000	4.20	0.008	1.11	0.00
0.30	0.585	0.407	1.000	4.54	0.007	1.14	0.00
0.35	0.585	0.406	1.000	4.54	0.008	1.15	0.00
0.40	0.591	0.416	1.000	4.30	0.008	1.15	0.00
0.45	0.591	0.416	1.000	4.27	0.008	1.16	0.00
0.50	0.611	0.445	1.000	3.94	0.008	1.17	0.00
0.55	0.644	0.492	1.000	3.36	0.008	1.18	0.00
0.60	0.643	0.490	1.000	3.36	0.008	1.19	0.00
0.65	0.643	0.490	1.000	3.35	0.008	1.19	0.00
0.70	0.643	0.489	1.000	3.35	0.008	1.20	0.00
0.75	0.642	0.488	1.000	3.35	0.008	1.21	0.00
0.80	0.574	0.487	0.777	3.36	0.008	1.22	0.33
0.85	0.563	0.474	0.771	3.66	0.008	1.23	0.33
0.90	0.561	0.473	0.766	3.66	0.008	1.24	0.33
0.95	0.554	0.465	0.761	3.71	0.014	1.25	0.33
1.00	0.552	0.464	0.756	3.71	0.014	1.26	0.33

Why the other DAS values lost

0.00–0.50 loses on localization. Mean distance of the top-ranked spot is 4.74 cm, against 3.36 cm for the winner. The energy is not locked on the labeled tumor.

0.60–0.75 sits on the same plateau as the winner (average score 0.643 vs 0.644). The gap is under 0.01, so these are not a separate result. The default is the highest point on that plateau.

0.80–1.00 raises healthy false positives to 0.33 against 0.00 for the winner.

DMAS pair exponent at the winning companions

Value	Average	Tumor	Healthy	Loc cm	Area	Ring	Healthy FP
0.00	0.590	0.414	1.000	4.54	0.007	1.01	0.00
0.05	0.554	0.402	0.906	5.38	0.008	1.04	0.00
0.10	0.587	0.451	0.903	4.15	0.008	1.04	0.00
0.15	0.586	0.452	0.900	4.16	0.008	1.04	0.00
0.20	0.588	0.455	0.897	4.08	0.008	1.04	0.00
0.25	0.585	0.453	0.894	4.28	0.008	1.05	0.00
0.30	0.618	0.501	0.892	3.24	0.008	1.05	0.00
0.35	0.610	0.490	0.889	3.37	0.008	1.05	0.00
0.40	0.604	0.483	0.886	3.41	0.008	1.06	0.00
0.45	0.596	0.473	0.883	3.43	0.008	1.06	0.00
0.50	0.591	0.467	0.881	3.63	0.008	1.06	0.00
0.55	0.588	0.464	0.878	3.71	0.008	1.06	0.00
0.60	0.608	0.493	0.876	3.39	0.008	1.06	0.00
0.65	0.610	0.497	0.874	3.32	0.008	1.06	0.00
0.70	0.608	0.496	0.871	3.34	0.008	1.07	0.00
0.75	0.602	0.487	0.869	3.34	0.008	1.07	0.00
0.80	0.587	0.467	0.867	3.34	0.008	1.07	0.00
0.85	0.582	0.460	0.866	3.32	0.008	1.07	0.00
0.90	0.571	0.445	0.864	3.32	0.008	1.07	0.00
0.95	0.581	0.461	0.863	3.33	0.008	1.08	0.00
1.00	0.565	0.438	0.861	3.90	0.008	1.08	0.00

Why the other pair exponents lost

0.00–0.25 loses on localization. Mean distance of the top-ranked spot is 4.43 cm, against 3.24 cm for the winner. The energy is not locked on the labeled tumor.

0.35 sits on the same plateau as the winner (average score 0.610 vs 0.618). The gap is under 0.01, so these are not a separate result. The default is the highest point on that plateau.

0.40–0.60 is simply lower on the combined average (0.597 vs 0.618). Localization 3.51 cm, area fraction 0.008. The loss is spread across the score, not one dramatic failure.

0.65–0.70 sits on the same plateau as the winner (average score 0.609 vs 0.618). The gap is under 0.01, so these are not a separate result. The default is the highest point on that plateau.

0.75–0.95 is simply lower on the combined average (0.585 vs 0.618). Localization 3.33 cm, area fraction 0.008. The loss is spread across the score, not one dramatic failure.

1.00 loses on localization. Mean distance of the top-ranked spot is 3.90 cm, against 3.24 cm for the winner. The energy is not locked on the labeled tumor.

DMAS coherence gamma at the winning companions

Value	Average	Tumor	Healthy	Loc cm	Area	Ring	Healthy FP
0.00	0.618	0.501	0.892	3.24	0.008	1.05	0.00
0.05	0.615	0.498	0.889	3.27	0.008	1.05	0.00
0.10	0.581	0.450	0.887	4.30	0.008	1.05	0.00
0.15	0.580	0.450	0.885	4.30	0.008	1.05	0.00
0.20	0.581	0.451	0.883	4.10	0.008	1.05	0.00
0.25	0.580	0.451	0.881	4.10	0.008	1.06	0.00
0.30	0.579	0.450	0.879	4.10	0.008	1.06	0.00
0.35	0.577	0.448	0.877	4.17	0.008	1.06	0.00
0.40	0.576	0.448	0.875	4.15	0.008	1.06	0.00
0.45	0.575	0.447	0.873	4.16	0.008	1.06	0.00
0.50	0.570	0.441	0.872	4.16	0.008	1.07	0.00
0.55	0.569	0.441	0.870	4.16	0.008	1.07	0.00
0.60	0.569	0.440	0.869	4.16	0.008	1.07	0.00
0.65	0.568	0.440	0.868	4.16	0.008	1.07	0.00
0.70	0.572	0.446	0.866	4.09	0.008	1.07	0.00
0.75	0.558	0.427	0.865	4.12	0.008	1.07	0.00
0.80	0.558	0.427	0.864	4.12	0.008	1.08	0.00
0.85	0.558	0.427	0.863	4.12	0.008	1.08	0.00
0.90	0.547	0.412	0.861	4.12	0.008	1.08	0.00
0.95	0.546	0.411	0.860	4.17	0.008	1.08	0.00
1.00	0.546	0.411	0.859	4.17	0.008	1.08	0.00

Why the other DMAS gamma values lost

0.05 sits on the same plateau as the winner (average score 0.615 vs 0.618). The gap is under 0.01, so these are not a separate result. The default is the highest point on that plateau.

0.10–1.00 loses on localization. Mean distance of the top-ranked spot is 4.16 cm, against 3.24 cm for the winner. The energy is not locked on the labeled tumor.

DMAS-D4 exponent at the winning companions

Value	Average	Tumor	Healthy	Loc cm	Area	Ring	Healthy FP
0.00	0.522	0.317	1.000	9.39	0.004	1.00	0.00
0.05	0.630	0.471	1.000	3.70	0.008	1.01	0.00
0.10	0.655	0.507	1.000	3.24	0.008	1.02	0.00
0.15	0.655	0.508	1.000	3.24	0.008	1.03	0.00
0.20	0.656	0.508	1.000	3.24	0.008	1.04	0.00
0.25	0.655	0.508	1.000	3.24	0.008	1.05	0.00
0.30	0.655	0.507	1.000	3.24	0.008	1.06	0.00
0.35	0.654	0.506	1.000	3.24	0.008	1.07	0.00
0.40	0.653	0.505	1.000	3.24	0.008	1.08	0.00
0.45	0.652	0.503	1.000	3.24	0.008	1.09	0.00
0.50	0.651	0.502	1.000	3.24	0.008	1.10	0.00
0.55	0.650	0.501	1.000	3.24	0.008	1.11	0.00
0.60	0.650	0.500	1.000	3.24	0.008	1.12	0.00
0.65	0.649	0.498	1.000	3.24	0.008	1.13	0.00
0.70	0.648	0.498	1.000	3.24	0.008	1.14	0.00
0.75	0.648	0.497	1.000	3.24	0.008	1.16	0.00
0.80	0.647	0.496	1.000	3.24	0.008	1.17	0.00
0.85	0.588	0.496	0.804	3.24	0.008	1.18	0.33
0.90	0.586	0.495	0.796	3.24	0.008	1.19	0.33
0.95	0.583	0.495	0.790	3.24	0.008	1.20	0.33
1.00	0.581	0.495	0.783	3.24	0.008	1.22	0.33

Why the other D4 exponents lost

0.00 loses on localization. Mean distance of the top-ranked spot is 9.39 cm, against 3.24 cm for the winner. The energy is not locked on the labeled tumor.

0.05 is simply lower on the combined average (0.630 vs 0.655). Localization 3.70 cm, area fraction 0.008. The loss is spread across the score, not one dramatic failure.

0.10–0.20 sits on the same plateau as the winner (average score 0.655 vs 0.655). The gap is under 0.01, so these are not a separate result. The default is the highest point on that plateau.

0.30–0.80 sits on the same plateau as the winner (average score 0.651 vs 0.655). The gap is under 0.01, so these are not a separate result. The default is the highest point on that plateau.

0.85–1.00 raises healthy false positives to 0.33 against 0.00 for the winner.

Per-scan localization of the blind top spot (cm). Healthy scans have no target distance.

Gen	Scan	Label	DAS winner	DAS textbook	D4 winner	D4 textbook
gen1	4	healthy	healthy	healthy	healthy	healthy
gen1	3	small/low/sphere/lower-right	2.35	1.67	2.19	2.19
gen1	7	medium/moderate/sphere/upper-left	6.73	9.82	6.67	6.67
gen1	12	large/high/sphere/upper-right	7.61	9.24	7.26	7.61
gen2	0	healthy	healthy	healthy	healthy	healthy
gen2	106	small/high/sphere/lower-left	2.98	6.27	2.48	4.59
gen2	104	medium/high/sphere/central	0.86	4.89	0.85	1.92
gen2	105	medium/high/sphere/upper-left	0.31	6.55	0.29	0.31
gen3	1	healthy	healthy	healthy	healthy	healthy
gen3	6	small/low/sphere/upper-right	3.05	3.28	3.05	3.05
gen3	16	medium/moderate/sphere/upper-left	3.18	3.02	3.32	3.32
gen3	48	medium/moderate/sphere/lower-right	3.18	3.29	3.05	3.05

The full narrative, including every named alternative (0.25, 0.35, 0.50, 0.55, 0.75), is in results/beamformer_constant_sweep_detailed.md. Plots are in results/beamformer_sweep/.